Math 629 Spring 26 **Name: Mengxiang Jiang**

Turn this assginment in on gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

1. Show that Fermat's two methods of determining a maximum or minimum of a polynomial p(x) are both equivalent to solving $p'(x) = 0$.

Adequality: Fermat observed that near a maximum or minimum, $p(x)$ changes very little. He replaced x with $x + e$ and set $p(x)$ adequate to $p(x + e)$:

$$p(x) \approx p(x + e).$$

Rearranging and dividing both sides by e:

$$\frac{p(x + e) - p(x)}{e} \approx 0.$$

Expanding $p(x+e)$ by the binomial theorem, every remaining term still contains a factor of e. Setting $e = 0$ to suppress the error gives precisely:

$$\lim_{e \to 0} \frac{p(x + e) - p(x)}{e} = p'(x) = 0.$$

Double Root: At an extremum x_0, the horizontal line $y = p(x_0)$ is tangent to the curve, so $p(x) = p(x_0)$ has a double root at x_0:

$$p(x) - p(x_0) = (x - x_0)^2, q(x)$$

for some polynomial $q(x)$. Differentiating both sides:

$$p'(x) = 2(x - x_0), q(x) + (x - x_0)^2, q'(x).$$

Evaluating at $x = x_0$:

$$p'(x_0) = 0.$$

Conversely, if $p'(x_0) = 0$, write $p(x) - p(x_0) = (x - x_0), r(x)$. Differentiating gives $p'(x_0) = r(x_0) = 0$, so $(x - x_0) \mid r(x)$, hence $(x - x_0)^2 \mid p(x) - p(x_0)$: the root is indeed double.

2. Use one of Fermat's methods to find the maximum of $bx - x^3$.

Replace x with $x + e$ and set $p(x)$ adequate to $p(x + e)$:

$$bx - x^3 \approx b(x + e) - (x + e)^3.$$

Expand the right side:

$$b(x + e) - (x + e)^3 = bx + be - x^3 - 3x^2e - 3xe^2 - e^3.$$

Subtract $bx - x^3$ from both sides:

$$0 \approx be - 3x^2e - 3xe^2 - e^3.$$

Divide through by e:

$$0 \approx b - 3x^2 - 3xe - e^2.$$

Setting $e = 0$:

$$b - 3x^2 = 0 \implies x = \sqrt{\frac{b}{3}}.$$

The maximum value is:

$$p\left(\sqrt{\tfrac{b}{3}}\right) = b\sqrt{\tfrac{b}{3}} - \left(\sqrt{\tfrac{b}{3}}\right)^3 = \frac{2b}{3}\sqrt{\frac{b}{3}}.$$

3. Modify Fermat's tangent method to be able to apply it to curves given by equations of the form $f(x, y) = c$. Begin by noting that if $(x + e, \bar{y})$ is a point on the tangent line near to (x, y), then $\bar{y} = \frac{t+e}{t}y$. Then adequate $f(x, y)$ to $f(x + e, \frac{t+e}{t}y)$.

$$f(x, y) \approx f\left(x + e,\ \frac{t + e}{t}y\right).$$

Expand the right side using the binomial theorem, collecting only terms of order e^1 and discarding higher powers:

$$f\left(x + e,\ \frac{t + e}{t}y\right) = f(x, y) + e \cdot A + \frac{ey}{t} \cdot B + O(e^2)$$

where A and B are the coefficients obtained by expanding in e. Subtracting $f(x, y)$, dividing by e, then setting $e = 0$ gives:

$$A + \frac{y}{t} \cdot B = 0 \implies t = -\frac{y \cdot B}{A}.$$

4. Show in modern notation, Fermat's method of finding the subtangent t to $y = f(x)$ determines t as $t = f(x)/f'(x)$.

   At a point (x, y) on $y = f(x)$, the tangent line meets the x-axis at a distance t from the foot of the ordinate. By similar triangles, the point $(x + e, \bar{y})$ on the tangent line has:

$$\bar{y} = \frac{t+e}{t} y = \frac{t+e}{t} f(x).$$

   Since $(x + e, \bar{y})$ is near the curve, adequate $f(x + e)$ to $\bar{y}$:

$$f(x+e) \approx \frac{t+e}{t} f(x).$$

   Rearranging:

$$f(x+e) - f(x) \approx \frac{e}{t} f(x).$$

   Dividing both sides by e:

$$\frac{f(x+e) - f(x)}{e} \approx \frac{f(x)}{t}.$$

   Setting $e = 0$ (suppressing the error), the left side becomes $f'(x)$:

$$f'(x) = \frac{f(x)}{t}.$$

   Solving for t:

$$t = \frac{f(x)}{f'(x)}.$$